

ITA0608 - MACHINE LEARNING LAB

PROGRAMS

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Experiment 7: Logistic

Regression (LR) Algorithm

AIM :

To implement the Logistic Regression (LR) algorithm in Python for binary classification using an appropriate dataset.

PROCEDURE :

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features.
5. Create the Logistic Regression model.
6. Train the model using the training data.
7. Predict the class labels for the test data.
8. Evaluate the model using accuracy and confusion matrix.
9. Display the prediction results.

ALGORITHM :

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Scale the input features.
5. Initialize the Logistic Regression model.
6. Train the model using the training data.
7. Predict the class labels for the test data.
8. Display the predicted results and accuracy.
9. Stop.

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OUTPUT :

RESULT :

The Logistic Regression (LR) algorithm was successfully implemented in Python. The model was trained and tested using the Breast Cancer dataset and achieved high classification accuracy for binary classification.